

Boyle, Robert, English chemist, physicist, and inventor (1627–91) See p.111

Brahe, Tycho, Danish astronomer (1546–1601) See p.87

Bramah, Joseph (1748–1814) English locksmith noted for inventing the hydraulic press, an improved water closet, a machine for printing bank notes, and a wood planing machine. He also built a pick-proof lock—a model of which was left as a challenge in his store window and remained unpicked for 67 years, despite numerous attempts.

Brewster, David (1781–1868) Scottish physicist best known for his work in optics, including polarization, reflection, refraction, and light absorption. The invention of the kaleidoscope and an improved stereoscope popularized Brewster's name and his portrait was displayed on cigar boxes.

Broca, Paul (1824–80) French surgeon who discovered the part of the frontal lobe responsible for articulate speech—now known as Broca's area. Broca found that lesions in this area of the brain caused aphasia, which impairs the ability to form articulate words. His studies of the brain helped establish physical anthropology.

Brunel, Isambard Kingdom (1806–59) English engineer whose bridges, railroad lines, and steamships revolutionized modern engineering. Brunel helped his father build the first tunnel under the Thames River, designed the Clifton Suspension Bridge across the Avon River, and constructed the Great Western Railroad from London to Cornwall. He also built three steamships, including the *Great Western*—the first regular transatlantic passenger ship.

Buffon, Georges (1707–88) French naturalist and mathematician best known for his 36-volume *Natural History* (1749–1788). After studying law, medicine, and mathematics, Buffon devoted himself to the study of natural history and was an early proponent of evolution.

Carnot, Nicolas Leonard Sadi (1796–1832) French physicist and military engineer often considered the father of thermodynamics. Carnot's 1824 *Reflections on the Driving Power of Fire* presented the Carnot cycle, which is now considered the most efficient heat engine allowed by physical laws. Carnot's work was largely ignored until after his death, when he was credited with introducing the second law of thermodynamics.

Carson, Rachel Louise, American marine biologist (1907–64) See p.290

Cassini, Giovanni Domenico (1625–1712) Italian-born French astronomer who discovered a dark gap between two of Saturn's rings, now called Cassini's division. He also discovered four of Saturn's moons and Jupiter's Great Red Spot. Cassini was the first to regard zodiacal light as a cosmic rather than a meteorological phenomenon.

Cauchy, Augustin-Louis, Baron (1789–1857) French mathematician, writer, and a pioneer of analysis. In five textbooks and over 800 research articles, he presented innovative research on infinitesimal calculus, probability, mathematical physics, and other subjects.

Cavendish, Henry (1731–1810) English physicist and chemist and noted for his

study of "inflammable air" (hydrogen). A wealthy recluse, Cavendish devoted his life to conducting scientific experiments on a wide range of topics, including chemistry, electricity, and a celebrated experiment to calculate the weight of Earth.

Celsius, Anders, Swedish astronomer (1701–44) See p.140

Chadwick, James (1891–1974) English physicist awarded the 1935 Nobel Prize in Physics for discovering the neutron, a particle without electric charge in the nucleus of an atom. He joined the Manhattan Project and was knighted in 1945.

Chandrasekhar, Subrahmanyan (1910–95) Indian-born American astrophysicist awarded the 1983 Nobel Prize in Physics for showing that white dwarf stars can exist only up to a maximum mass, the Chandrasekhar limit, of about 1.44 times that of the Sun. Initially rejected, this later helped the understanding of neutron stars, supernovas, and black holes.

Chambers, Robert (1802–71) Scottish publisher and anonymous writer of the enormously controversial *Vestiges of the Natural History of Creation* (1844). Only acknowledged posthumously as the book's author, Chambers wrote several other historical, literary, and geological titles and published the *Edinburgh Journal* and *Chambers' Encyclopaedia*.

Chappe, Claude (1763–1805) French engineer who invented a mechanical semaphore system to connect the French mainland and bring news of Napoleon Bonaparte's campaign. In 1772, Chappe and his brother Ince successfully delivered their first message between Paris and Lille. By 1774, 513 semaphore towers spanned France and parts of Europe. Chappe's system was superseded by the electric telegraph in 1846.

Chargaff, Erwin, Austrian biochemist (1905–2002) See p.279

Châtelet, Émilie du (1706–49) French physicist and mathematician noted for her translation of Isaac Newton's *Principia Mathematica*, still the only complete one. Living with her lover Voltaire, she wrote several important books on science, philosophy, and religion.

Cherenkov, Pavel Alekseyevich (1904–90) Russian physicist who shared the 1958 Nobel Prize in Physics with Igor Tamm and Ilya Frank for discovering Cherenkov radiation. Cherenkov observed that electrons emit a blue glow when traveling through a medium such as water at speeds faster than light in that medium. Based on this effect, the Cherenkov detector is used in experimental nuclear and particle physics.

Cohen, Stanley Norman (1935–) American geneticist, microbiologist, and early pioneer of genetic engineering. From 1972, Cohen collaborated with Stanford University colleagues Herbert Boyer and Paul Berg to combine and transplant genes. This led to the first genetic engineering experiment, which transferred frog ribosomal RNA into bacteria cells.

Cope, Edward Drinker (1840–97) Pioneering American paleontologist whose discovery of over 1,000 extinct Tertiary Period vertebrates helped define modern paleontology. Cope wrote over 1,200 papers on his finds, which included extinct fish and dinosaurs. From 1877, Cope competed with rival Othniel Marsh

in the "bone wars" to discover the greatest number of fossils, damaging the reputation and finances of both men.

Copernicus, Nicolaus, Polish astronomer (1473–1543) See p.76

Coriolis, Gaspard-Gustave de (1792–1843) French engineer and mathematician best known for the Coriolis force, which affects movement across a rotating body, such as air masses around Earth. Coriolis dedicated his life to studying applied mechanics, friction, and hydraulics, and introduced the terms "work" and "kinetic energy" into scientific parlance.

Crick, Francis (1916–2004) British biophysicist and neuroscientist who determined the structure of deoxyribonucleic acid, or DNA, with colleague James Watson. Their discovery confirmed that DNA contained life's hereditary information and earned Crick, Watson, and biophysicist Maurice Wilkins the 1962 Nobel Prize in Physiology or Medicine.

Crookes, William (1832–1919) British chemist and physicist noted for pioneering vacuum tubes and discovering the element thallium. After inheriting a fortune, he devoted himself to scientific research, inventing the Crookes tube to investigate cathode rays, founding the journal *Chemical News*, and inventing the radiometer to convert light radiation into rotary motion.

Curie, Marie, Polish–French physicist and chemist (1867–1934) See p.233

Cuvier, Georges (1769–1832) French zoologist who established comparative anatomy and paleontology by comparing fossils with living animals. His studies proved that whole species of creatures had become extinct. He attributed mass extinctions to extreme catastrophic events, a theory known as catastrophism.

Da Vinci, Leonardo, Italian artist, architect, botanist, mathematician, and engineer (1452–1519) See p.71

Daguerre, Louis (1787–1851) French painter and physicist who perfected a process of creating permanent photographs on thin copper sheets, called daguerreotypes.

Dalton, John, British chemist and physicist (1766–1844) See p.172

Darwin, Charles, British naturalist (1809–82) See p.206

Darwin, Erasmus (1731–1802) English physician, poet, and inventor, and grandfather of naturalist Charles Darwin. Darwin was a prominent figure best known for his scientific poetry, freethinking ideas, and mechanical inventions. His *Zoonomia* outlined his radical theories on evolution.

Davy, Humphry (1778–1829) English chemist and pioneer of electrochemistry, noted for using electrolysis to isolate and discover several chemical elements, including sodium, potassium, barium, and magnesium. He also invented the Davy gas safety lamp for miners and was made a baronet in 1818.

Dawkins, Richard, British zoologist and evolutionary biologist (1941–) See p.307

Delbrück, Max (1906–81) German-born US biophysicist and pioneer of molecular biology. Trained in physics, Delbrück began working

on chemistry after fleeing Nazi Germany for America in 1937. He was awarded the 1969 Nobel Prize in Physiology or Medicine with Alfred Day Hershey and Salvador Luria for their work on bacteriophages—viruses that infect bacteria and then replicate.

Descartes, René (1596–1650) French mathematician known as the father of modern philosophy. His principle "I think, therefore I am" summarizes his determination to build only on knowledge that is certain. He also founded Cartesian geometry and contributed to optics.

Diesel, Rudolf (1858–1913) German engineer famous for inventing the diesel engine, a four-stroke, vertical cylinder compression engine that made him rich. Later, he disappeared from the deck of a channel steamer, and was presumed drowned.

Diophantus of Alexandria (c.200–c.284) Greek mathematician based in Alexandria and reputed as a founding father of algebra. His only surviving work is *Arithmetica*, the earliest known treatise on algebra, which greatly influenced Islamic scholars and also the French mathematician Pierre de Fermat, who helped found modern number theory.

Dirac, Paul, British theoretical physicist (1902–84) See p.262

Dollond, John (1706–61) English optician and manufacturer of astronomical instruments. Born to Huguenot silk weavers, Dollond is best known for reducing color distortion with achromatic lenses. He also invented the heliometer, a telescope used to measure the distances between stars.

Doppler, Christian Johann (1803–53) Austrian mathematician and physicist best known for the Doppler effect, which describes how the perceived frequency of light and sound waves produced by a moving source depends on the position of the observer. In 1850, he became professor of experimental physics at the University of Vienna.

Duchenne, Guillaume (1806–75) French neurologist who studied nervous and muscular disorders and developed electrotherapy to treat diseased nerves and atrophied muscles. Duchenne became the first to use deep tissue biopsy, clinical photography, and nerve conduction tests.

Eddington, Arthur, British astronomer, mathematician, and astrophysicist (1882–1944) See p.257

Edison, Thomas Alva, American inventor (1847–1931) See p.221

Ehrlich, Paul, German bacteriologist (1854–1915) See p.247

Einstein, Albert, German-born American physicist (1879–1955) See p.242

Eratosthenes (c.276–c.194 BCE) Greek mathematician and astronomer who first calculated the circumference of Earth. He was chief librarian at Alexandria in Egypt, where he measured the tilt of Earth's axis and calculated its circumference to be 250,000 stadia. Although the value of stadia is uncertain, his estimate was within the current range. He also created a calendar that included leap years, and created a system of latitude and longitude.